ACOUSTIC SPECIFICATION

Stanwell New Office

Prepared for Brian Hooper Architect

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Document Information

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1 Introduction

Northrop Consulting Engineers (Northrop) has been engaged by Brian Hooper Architect to provide acoustic advice for the Stanwell New Projects Office development, located on Switchyard Road, Stanwell (the Site). This Specification addresses the following:

- Determines relevant criteria and acoustic requirements for the development
- Outlines the modelling and calculations undertaken
- Provides recommendations to achieve target acoustic performance criteria

The following advice should be incorporated into the Detailed Design stage.

1.1. Application of this Specification

This Specification shall be read in combination with all other contract and design documentation for the project. This Specification captures the design intent at time of writing for the purpose and level of documentation requested. Any discrepancies between the other contract and design documentation and this Specification shall be clarified by the acoustic consultant.

The advice provided in this Specification is intended to capture the target acoustic performance. The buildability and installed performance of the construction and mitigation measures are the responsibility of the contractor.

Changes to the design shall be expected during the construction stage due to the nature of construction, site constraints and complexities encountered throughout the construction process.

This Specification should be used as an estimate for costing purposes only, not absolute. The principal contractor and sub-contractors shall allow suitable cost contingencies based on their experience.

1.2. Safety of Building Products

In accordance with the Amendments to the Queensland Building and Construction Commission and Other Legislation Amendment Act 2004, any building products specified or used shall be verified by the contractor as being safe and appropriate for use. Northrop does not take any responsibility for the use of unsafe products.

1.3. Referenced Documents

This assessment has been prepared considering the following documentation:

1.3.1. PROJECT DOCUMENTS

- Architectural drawings issued by Brian Hooper Architect/ m3architecture (see Appendix A)
- Return Brief issued by Northrop (ref: IB250172-AU01-BR01-1 Stanwell New Office Return Brief dated 14/07/2025)

1.3.2. CONSENT AUTHORITY, DESIGN GUIDELINES AND STANDARRDS

- Australian / New Zealand Standard 2107:2016 Acoustics – Recommended design sound levels and reverberation times for building interiors – issued by Standards Australia
- Australian Standard 2822:1985 Acoustics – Methods of assessing and predicting speech privacy and speech intelligibility – issued by Standards Australia

1.4. Project Understanding and Scope

The development is proposed to be an office building comprising of the following rooms / areas:

- 7 meeting rooms
- Training room
- Boardroom
- Health room

- Open area office
- Private office
- Outdoor garden area
- Covered outdoor
- Kitchen
- Toilets
- Comms room
- Store room
- First aid room

Northrop have been engaged to provide acoustic advice for input into design documentation pertaining to:
- Internal building services noise control
- Reverberation control
- Acoustic separation between spaces (airborne)
- Environmental noise ingress

1.5. Site Description

The project is located on Switchyard Road, Stanwell. The Site and the approximate building location are shown in Figure 1 below.

Figure 1: Aerial view of site and approximate building location (source: QGlobe)

1.6. Existing Environment

Attended noise measurements were undertaken to determine the existing noise impacts on the new office building. Measurements were undertaken in line with the worst affected façade of the new office building (ie. the southern façade). A noise level of 69 dBA L_{eq} was measured at the worst affected façade. The predominant noise source was from industrial fans located in the buildings to the south. It was noted that the noise sources had low frequency content, which was considered in the design of the new office building.

2 Project Acoustic Criteria

2.1. Internal Design Sound Levels

Australian / New Zealand Standard 2107:2016 "Acoustics – Recommended design sound levels and reverberation times for building interiors" provides recommended design sound levels for different areas of occupancy in buildings. The recommended levels are given in terms of an equivalent continuous A-weighted noise level (L_{eq}). The recommended values for internal background noise levels are shown in Table 1 below.

Table 1: AS/NZS 2107 – recommended design sound level ranges

TYPE OF OCCUPANCY/ACTIVITY	RECOMMENDED DESIGN SOUND LEVEL RANGE – L_{eq} dBA
Open plan office	40 – 45
Meeting rooms	40 – 45
General office areas	40 – 45
Board and conference rooms	30 – 40
Quiet rooms	40 – 45
Cafeterias	45 – 50
Toilets	45 – 55

2.2. Reverberation

Australian / New Zealand Standard 2107:2016 "Acoustics – Recommended design sound levels and reverberation times for building interiors" provides recommended reverberation times for different areas of occupancy in buildings. Reverberation times are given as the arithmetic average of 500 Hz – 1 kHz octave bands. The recommended values for reverberation times are shown in Table 2 below.

Table 2: AS/NZS 2107 – recommended reverberation time ranges

TYPE OF OCCUPANCY/ACTIVITY	RECOMMENDED REVERBERATION TIME RANGE – RT60 (S)
Meeting rooms (Small)	< 0.6
General office areas (including open plan)	0.4 – 0.6
Board and conference rooms	0.6 – 0.8
Quiet rooms	< 0.6
Cafeterias	< 1.0
Toilets	-

As well as overall reverberation time, impacts of acoustic defects such as flutter echoes and late reflections which can affect acoustic comfort and speech intelligibility should be considered. Placement of acoustically absorptive and reflective finishes should be coordinated to ensure these effects are minimised in sensitive rooms.

2.3. Internal Airborne Acoustic Insulation

There are no specific regulatory requirements or standards for internal wall acoustic insulation performance for office buildings. In the absence of specific requirements, best practice shall be applied to consider the client's requirements, the sensitivity and speech privacy requirements of each occupied room and the anticipated noise generated in adjacent areas to determine the acoustic insulation performance of internal walls, doors, internal glazing, and operable walls.

Internal airborne acoustic insulation requirements are quantified by a weighted sound reduction index (R_w) rating, which represents the design value of the proposed construction. Note that R_w is determined in a laboratory - a highly controlled environment - and should only be used as an indicative value for design purposes. Weighted standardised level difference (D_w) is a more robust measure of airborne sound insulation that is used for field measurements so includes doors, seals, mechanical grilles and associated flanking paths.

2.3.1. SPEECH PRIVACY

The level of speech privacy between adjacent rooms is dependent on many factors including the background noise level, vocal effort, reverberation within the room and the acoustic insulation performance of any walls, doors, internal glazing, operable walls, and ceilings separating the two spaces.

AS2822-1985 provides methods for assessing and predicting speech privacy. To assist in the determination of the acoustic separation requirements, speech privacy has been categorised into the following subjective levels presented in Table 3.

Table 3: Subjective speech privacy categories

PRIVACY CATEGORY	SUBJECTIVE DESCRIPTION	TYPICAL ROOM TYPES
Low/no privacy	Speech is understood and loud speech may be a source of distraction.	Storerooms, open plan areas, breakout areas
Normal	Conversational speech can be heard but not well understood.	Classrooms, shared office, informal meeting room
Raised	Raised speech can be heard but not well understood. Conversational speech is heard as a murmur.	Private offices and meeting rooms
Confidential	High levels of privacy for very private/sensitive discussions. Awareness of discussions in adjoining rooms is very low.	Conference and board rooms

2.3.2. ACOUSTIC SEPARATION CRITERIA

The wall mark-up in Appendix B provides the recommended/target internal wall acoustic airborne insulation performance for each area given the uses of the space and adjacent areas.

2.3.2.1. Internal Walls

The recommended R_w for walls not containing doors, glazing or operable walls dependent on activity noise and noise tolerance in adjacent rooms is presented in Table 4 below.

Table 4: Target wall R_w

TARGET WALL R_w (NOT INCLUDING DOORS, GLAZING OR OPERABLE WALLS)		EXPECTED SOURCE ROOM NOISE GENERATION			
		LOW	AVERAGE	HIGH	VERY HIGH
Receiver room sensitivity / privacy level	Low sensitivity / no privacy	–	–	40[1]	45[1]
	Medium sensitivity / normal privacy	40	42	45	48
	High sensitivity / raised privacy	42	45	48	50

TARGET WALL R_w (NOT INCLUDING DOORS, GLAZING OR OPERABLE WALLS)		EXPECTED SOURCE ROOM NOISE GENERATION			
		LOW	AVERAGE	HIGH	VERY HIGH
	Very high sensitivity / confidential privacy	45	48	50	55

1. Habitable rooms only

To minimise flanking paths, it is recommended that acoustic walls extend to full height, with penetrations to be fully sealed to an approved detail and acoustic sealant. Transfer ducts should be avoided for spaces that require speech privacy.

For sensitive areas adjacent to vibration or impact generating equipment, it is recommended that walls are "discontinuous" construction: minimum 20 mm cavity between 2 separate leaves, and for masonry, where wall ties are required to connect leaves, the ties are of the resilient type; for other than masonry, there is no mechanical linkage between leaves except at the periphery.

2.3.2.2. Internal Doors

Doors, due to their nature, limit the acoustic insulation performance of a partition. Acoustic doors are generally solid core construction with perimeter and threshold seals.

It is recommended that doors in acoustic walls are acoustically treated to target minimum R_w 30 – 35 for speech privacy and noise control. For critical applications such as confidential privacy requirements adjacent to high traffic areas or high noise generating areas, higher rated proprietary systems and/or airlocks are recommended.

The following is generally recommended for doors in acoustic walls:

- Strategic planning of door locations – e.g. locating doors to sensitive rooms as far from noisy rooms as much as possible; offsetting doors to sensitive rooms across a common corridor or circulation area; and maximizing the distance between doors to adjacent sensitive rooms
- Sliding doors in acoustic walls are recommended to be acoustically rated and be proprietary systems from specialist suppliers
- Consideration of air locks for sensitive rooms
- Door grilles are not appropriate in acoustic doors
- Installation of acoustic treatments including solid core construction, perimeter and threshold seals, acoustically rated frames
- Avoid pivot doors – these cannot be sufficiently sealed and therefore acoustic performance is limited

2.3.2.3. Internal Glazing

Glazing generally limits the acoustic insulation performance of solid partition walls. Glazed sections of partitions are recommended to be minimised in acoustic walls. If glazing is to be used in acoustic walls, the following is recommended:

- Target acoustic insulation performance of R_w 45 – 51 for speech privacy and noise control
- Use acoustically rated frames and seals
- Generally, laminated glass provides higher acoustic insulation than monolithic glass and should be considered
- Glazing is not recommended in walls which require confidential speech privacy or walls separating high noise areas with sensitive uses.

3 Construction Recommendations

3.1. Building Envelope

The building envelope should be upgraded to reduce the external industrial noise to the internal levels specified in Section 2.1. Table 5 summarises the acoustic recommendations that have been provided for the building envelope.

Table 5: Building envelope acoustic recommendations

ELEMENT	CONSTRUCTION	ACOUSTIC RATING (dB)
External glazing (including courtyard glazing)	10.38mm laminated glass or double glazed unit consisting of 10mm glass, 12mm air gap, 6mm glass	R_w 36
External wall (above ceiling line)	Profiled metal cladding, 35mm girt, 6mm RAB, 150mm steel stud with insulation (fibreglass 10kg/m^3)	R_w 33 (Ctr -6)
External wall (below ceiling line)	Profiled metal cladding, 35mm girt, 6mm RAB, 150mm steel stud with insulation (fibreglass 10kg/m^3), 16mm furring channel and 9mm fibre cement	R_w 52 (Ctr -8)
Roof	0.42mm BMT roof metal sheet, 200mm gap using an ashgrid spacer and 90x45 batten, 75mm fibreglass insulation (10kg/m^3) and 1 layer of 17.5mm plywood (560kg/m^3)	R_w 38 (Ctr -8)
Floor	Floor substrate constructed from 19mm compressed fibre cement (1622kg/m^3)	R_w 32 (Ctr -2)

3.2. Acoustic Separation – Internal Partitions

3.2.1. WALLS – RECOMMENDED CONSTRUCTIONS

Table 6 shows acoustic partition recommendations for airborne sound insulation corresponding to the recommended airborne acoustic insulation performance shown in Appendix B. In addition, the following is recommended to ensure required acoustic performance is achieved:

- To minimise flanking paths, it is recommended that acoustic walls extend to full height, with penetrations to be fully sealed to an approved detail and acoustic sealant.
- Full height walls should be coordinated with any in-ceiling building services reticulation to ensure cross talk is minimised between sensitive areas.

Table 6: Recommended light-weight wall constructions for airborne sound insulation

TARGET PERFORMANCE	WALL CONSTRUCTION (SLAB TO SLAB)
R_w 51	9mm fibre cement, steel stud (0.55mm), cavity width 92mm with 75mm fibreglass (10kg/m^3) and 9mm fibre cement
R_w 52	9mm fibre cement, steel stud (0.55mm), cavity width 150mm with 75mm fibreglass (10kg/m^3) and 9mm fibre cement
R_w 35	2 layers of 16mm fire rated plasterboard (roof cavity wall)

Provided that the walls shown in Table 6 are full height (floor to roof), the acoustic separation ratings detailed in Appendix B will be satisfied.

It is noted that full height walls are not possible in all areas. Partial height walls are highlighted in Appendix B, and the recommended construction of the partial height wall is summarised in Table 7. The table below also requires the meeting room and boardroom ceilings to be constructed from solid plasterboard to minimise noise transfer between adjacent spaces. The recommended ceiling constructions are marked up in Figure 2. All connections and gaps are to be sealed with an acoustic sealant as specified in Section 3.5.

Table 7: Partial height wall construction

WALL ID	RECOMMEND PARTITION CEILING CONSTRUCTION	INDICATIVE ACOUSTIC RATING FOR PARTITION
Type 1	• 16mm fire rated plasterboard ceiling on either side of the partition • Insulation installed above the plasterboard ceiling (75mm Bradford Acoustigard or equivalent). Extent should cover most of the ceiling and above the partition • The partition should extend 100mm above the ceiling line	Rw 50
Type 2	• Perforated plasterboard on one side of the partition and a solid 13mm standard plasterboard on the other side • Insulation installed above the plasterboard ceilings (75mm Bradford Acoustigard or equivalent). Extent should cover most of the ceiling • Install Autex BaffleBlock (or acoustic equivalent) above the partition and compressed by 35%. BaffleBlock width to be minimum 600mm • The partition should extend 100mm above the ceiling line	Rw 40[1]
Type 3	• Perforated plasterboard on one side of the partition and a solid 13mm standard plasterboard on the other side • Insulation installed above the plasterboard ceilings (75mm Bradford Acoustigard or equivalent). Extent should cover most of the ceiling • The partition should extend to the underside of the plant area floor (16mm CFC) • Plant room wall in the ceiling cavity should be constructed from 2 layers of 16mm fire rated plasterboard	Rw 40
Type 4	• 13mm standard plasterboard on either side of the partition • Insulation installed above the plasterboard ceilings (75mm Bradford Acoustigard or equivalent). Extent should cover most of the ceiling • The partition should extend to the underside of the plant area floor (16mm CFC)	Rw 40
Type 5	• 13mm standard plasterboard on either side of the partition • Insulation installed above the plasterboard ceiling (75mm Bradford Acoustigard or equivalent). Extent should cover most of the ceiling and above the partition • The partition should extend 100mm above the ceiling line	Rw 40
Type 6	• Perforated plasterboard on one side of the partition and a solid 16mm standard plasterboard on the other side • Insulation installed above the plasterboard ceilings (75mm Bradford Acoustigard or equivalent). Extent should cover most of the ceiling • Install Autex BaffleBlock (or acoustic equivalent) above the partition and compressed by 35%. BaffleBlock width to be minimum 600mm • The partition should extend 100mm above the ceiling line	Rw 40[1]
Type 7	• 16mm standard plasterboard on either side of the partition • Insulation installed above the plasterboard ceiling (75mm Bradford Acoustigard or equivalent). Extent should cover most of the ceiling and above the partition • The partition should extend 100mm above the ceiling line	Rw 40

1. No test data is available for this specific construction. Verification measurements are recommended to confirm if the acoustic performance is fit for purpose

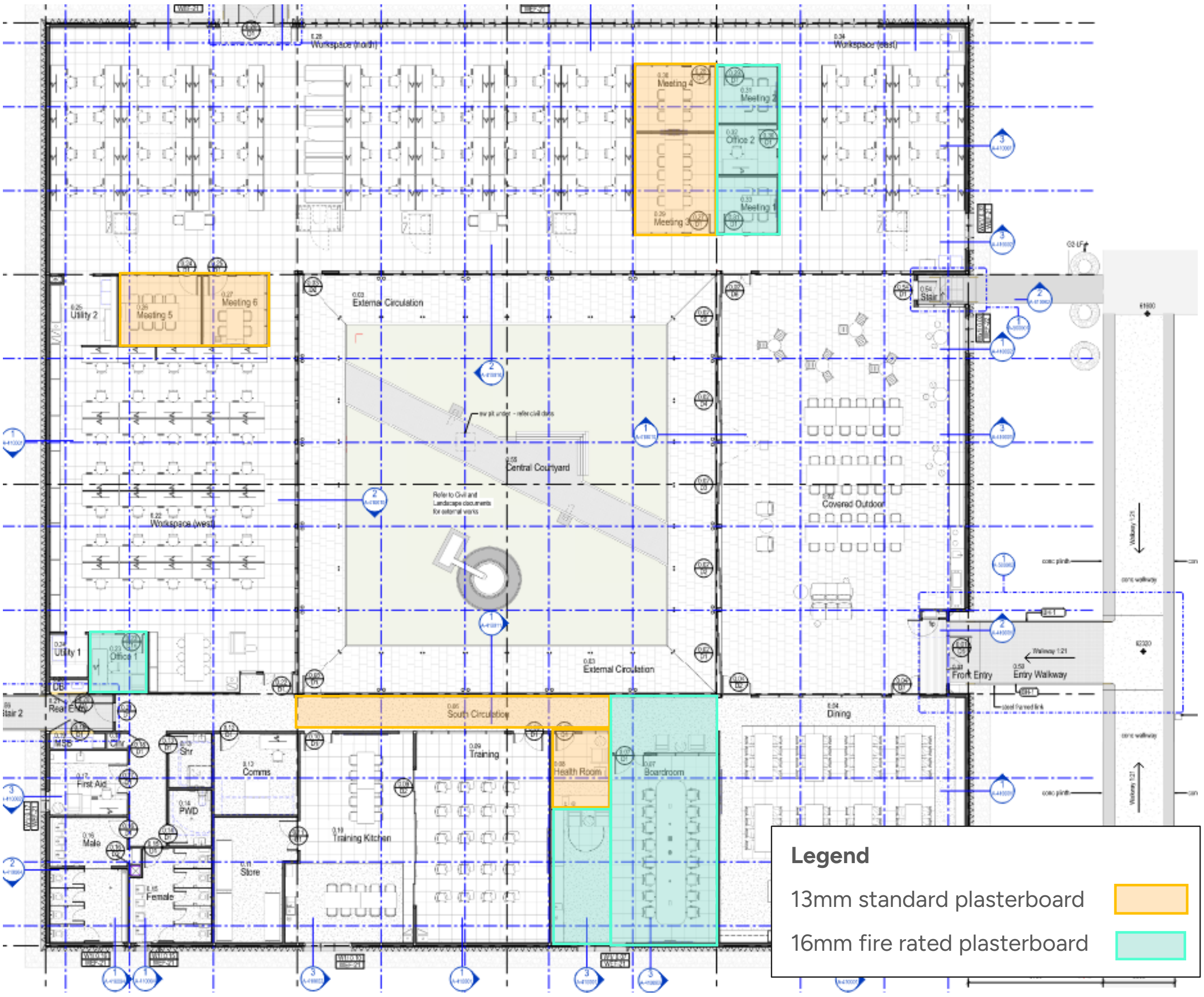

Figure 2: Solid ceiling upgrade mark up

3.2.2. DOORS

Table 8 below summarises acoustic door provisions corresponding to the recommended airborne acoustic insulation performance shown in the wall mark-up. In addition, the following is recommended to ensure required acoustic performance is achieved:

- Door framing and seals shall be acoustically rated so they do not degrade the performance of the partition.
- The contractor shall ensure that the installed doors are effectively sealed when closed.
- Pivot doors are not recommended in acoustically rated walls.
- Door assemblies shall include a threshold detail to ensure effective operation of bottom door seals including drop seals
- Door details shall be coordinated with non-acoustic requirements such as structure, fire, and security

Table 8: Recommended door constructions for airborne sound insulation

MARK-UP	DOOR TYPE	RECOMMENDED DOOR		RECOMMENDED SEALS	R_W (dB)
		TIMBER	GLASS		
1	Hinged with acoustic frames	Solid core 40 mm timber	10.38 mm laminated glass	Perimeter seal: Raven RP87Si, RP24, RP120 or equivalent Bottom seal: Raven RP38, RP99 or Lorient IS 8090 Si seals or equivalent	30

MARK-UP	DOOR TYPE	RECOMMENDED DOOR		RECOMMENDED SEALS	R_W (dB)
		TIMBER	GLASS		
2	Hinged with acoustic frames	Solid core 44 mm timber	12.38 mm laminated glass	Perimeter seal: Raven RP87Si, RP24, RP120 or equivalent Bottom seal: Raven RP38, RP99 or Lorient IS 8090 Si seals or equivalent	35
	Sliding	Lotus Glazed SL80 (or equivalent)			30

3.2.3. GLAZING

Table 9 below summarises internal glazing acoustic provisions corresponding to the recommended airborne acoustic insulation performance shown in the wall mark-up (Appendix B). Glass frames and seals are recommended to be acoustically rated such that they do not degrade the performance of the partition as a whole. Glazing is not recommended in walls that require confidential privacy or are adjacent to high noise generating areas.

Table 9: Recommended internal glazing for airborne sound insulation

MARK-UP	RECOMMENDED GLAZING	R_W (dB)
	10.38mm laminated glass	36
	12.5mm VLam Hush	40
	Double glazing consisting of 1 layer 12.5mm VLam Hush, 60mm air gap and another layer of 12.5mm VLam Hush	50

3.2.4. PLANT ROOM ACCESS DOORS

The ceiling access doors within rooms that require acoustic separation are to be acoustically treated. The minimum acoustic rating for the access door is to be Rw 31 (Trafalgar Rw 31C or equivalent).

3.3. Reverberation Control

Table 10 below summarises the recommendations for reverberation control treatment.

The recommendations below include consideration of flutter echoes and late reflections. For the purpose of mitigating these effects, calculated reverberation times may be lower than the recommended AS/NZS 2107 reverberation times are proposed, but based on measurements of similar spaces are suitable for this application.

The recommended upgrades are detailed in Table 10.

Table 10: Recommended reverberation control treatment

ROOM	REVERBERATION TIME TARGET (s)	PROPOSED FINISHES / ACOUSTIC TREATMENT REQUIRED / MINIMUM NRC OF TREATMENT
Dining/ kitchen	< 1	• Ceiling: perforated plasterboard with 75mm Bradford Acoustigard (14kg/m^3) • Wall: fibre cement sheet and glazing • Floor: cork flooring/ carpet
Boardroom	0.6 – 0.8	• Ceiling: Solid plasterboard

ROOM	REVERBERATION TIME TARGET (s)	PROPOSED FINISHES / ACOUSTIC TREATMENT REQUIRED / MINIMUM NRC OF TREATMENT
		• Wall: fibre cement sheet and glazing • Acoustic lining: Autex Cube 12mm (NRC 0.45). Minimum area $60m^2$ • Floor: cork flooring/ carpet
Training	0.4 – 0.6	• Ceiling: perforated plasterboard with 75mm Bradford Acoustigard ($14kg/m^3$) • Wall: fibre cement sheet and glazing • Floor: carpet
Training kitchen	0.4 – 0.6	• Ceiling: perforated plasterboard with 75mm Bradford Acoustigard ($14kg/m^3$) • Wall: fibre cement sheet and glazing • Floor: cork flooring/ carpet
Workspace (open plan)	0.4 – 0.6	• Ceiling: perforated plasterboard with 75mm Bradford Acoustigard ($14kg/m^3$) • Wall: fibre cement sheet and glazing • Floor: carpet
Meeting rooms	< 0.6	• Ceiling: Solid plasterboard • Wall: fibre cement sheet and glazing • Floor: carpet • Acoustic lining: Autex Cube 12mm (NRC 0.45). Refer to architectural drawings for acoustic lining locations.

3.4. Central Courtyard

The central courtyard is situated in the middle of the development surrounded by walls, resulting in partial shielding from the power plant noise. However, due to it being open, the outdoor courtyard will have a higher risk of high levels of noise exposure from the nearby power plant.

There may be an opportunity to mask the power plant noise in the courtyard using a sound masking speaker system or using a water feature to generate localised waterfall/ trickling noise.

3.5. Acoustic Sealing Methodology

Sealing materials fall into three classes – plasterboard setting compound, acoustic sealing (caulking) compound and cement or plaster render.

- Plasterboard setting compound shall only be used to seal joints in the exposed sheet of plasterboard in areas where the plasterboard is to be painted (for example, in corridors). It shall not be used for any other purpose. The reason for this is to ensure that cracks do not appear in the plasterboard construction as a result of movement in the structure.
- Acoustic sealing (caulking) compound is otherwise used in all other situations but shall only be applied where the gap is less than 5 mm wide. In such cases, the compound shall be applied to an equal depth as the width. For example, the gap between plasterboard and soffit slab should be no greater than 5 mm.
- Cement or plaster render: If the slab is uneven and results in a gap greater than 5 mm in parts, the slab shall be faced with cement or plaster render in the area concerned to ensure the gap is made even and is within specification.

Approved acoustic sealing (caulking) compounds are shown in Table 11.

Table 11: Approved acoustic sealing (caulking compounds)

MANUFACTURER	PRODUCT NAME	SPECIFIC GRAVITY
Bostik	Fireban 1	1.46 sg

MANUFACTURER	PRODUCT NAME	SPECIFIC GRAVITY
Sika	Firerate	1.5 sg
Promat Fyerguard	Promoseal Mastic	1.6 sg
CSR	Gyprock Fire Mastic	1.6 sg
Ramset	Blaze Brake 201	1.5 sg
Hilti	CP 606 Firestop and Acoustic Sealant	1.5 sg

Alternatively, provided that the acoustic caulking specific gravity is a minimum 1.46 sg, alternative acoustic sealing compound specifications and sample shall be submitted to the acoustic consultant for approval.

The building contractor shall use approved backing rods where necessary to ensure caulking compound is supported from behind.

In the case of A/C duct penetrations through walls and slabs, a tolerance of 5 mm gap may not be workable and a larger gap results. In this case, the gap shall be first packed tight with fire rated wool or equivalent. Caulking compound shall then be applied to the whole gap on both sides of the penetration to provide adhesion of metal to slab or wall. Sealing shall be applied over the whole girth of the duct and to a depth equal to the width of the gap. The gap shall then be fully sealed with a 3 mm thick steel angle fixed to both duct and slab or wall using caulking sealer along the whole length of the steel angle to ensure that small gaps are sealed closed. The width of the steel angle shall be sized according to the gap width to ensure the gap is fully covered.

Penetrations shall be sized for pipes passing through building slabs or walls to allow a uniform clearance of 10-15 mm around the item and this space shall be packed through its depth with fire rated wool or equivalent. The joints on both sides shall be sealed using an approved acoustic sealant 12 mm minimum depth.

Render can be plaster or cement based. This material shall be used to reduce uneven gaps to the specified maximum noted above. Specific applications may include the levelling of floor slabs and slab soffits.

The extent of rendering shall extend over an area sufficient to ensure a level base surface.

3.6. Services Penetrations

The following is recommended for building services penetrations to ensure required acoustic performance of the penetrated wall is not degraded:

- Electrical cables sealed with acoustic caulk, to ensure removal of all air gaps.
- All hydraulic pipes sealed with acoustic caulk, to ensure removal of all air gaps.
- Mechanical ducts sealed as per the detail shown in Figure 3, below. Flexible ductwork penetrations to acoustic full height walls are not recommended, if required, a section of rigid duct should be installed for the penetration to the detail below.
- Figure 3 is also applicable to penetrations through the plant floors above the meeting room ceilings

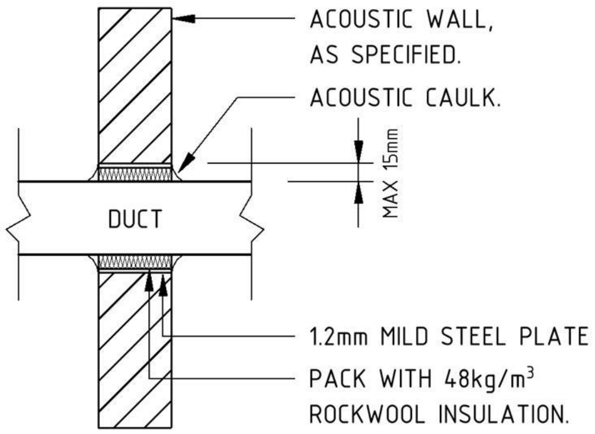

Figure 3: Duct penetration detail

3.7. Internal Mechanical Services Noise Assessment

A review of the proposed mechanical design has been undertaken based on the layouts and equipment selections.

The mechanical sub-contractor shall ensure that generated noise within the air distribution system does not exceed internal sound levels shown in Table 1.

The sub-contractor shall ensure that the airborne noise generated by air turbulence, air leakage, air moving equipment or by any other mechanical services equipment, is transmitted through air ducts, clearance holes etc. is minimised.

Allowance for the following acoustic treatments shall be made by the mechanical sub-contractor. Internal lining may be polyester or glasswool such as Bradford Supertel or equal approved.

3.7.1. INTERNAL MECHANICAL PLANT NOISE SOURCES

Table 12 outlines the sound power levels of each of the assessed units affecting the internal areas.

Table 12: Sound power levels of mechanical plant sources towards internal rooms

Type	Unit Designation	Manufacturer	Model	Unit / Room Servicing	Number Of Units	Sound Power Level per unit, dBA
AC	AC-1-1	Daikin	FXMQ180PV1A	Workspace (north)	1	86
	AC-1-2	Daikin	FXMQ50PAVE	Meeting room 2, meeting room 4, private office	1	83
	AC-1-3	Daikin	FXMQ250PV1A	Workspace (north), workspace (east)	1	86
	AC-1-4	Daikin	FXMQ100PAVE	Workspace (east)	1	81
	AC-1-5	Daikin	FXMQ50PAVE	Meeting room 3	1	83
	AC-1-6	Daikin	FXMQ25PAVE	Meeting room 1	1	71
	AC-2-1	Daikin	FXMQ125PAVE	Workspace (north), workspace (west)	1	82

Type	Unit Designation	Manufacturer	Model	Unit / Room Servicing	Number Of Units	Sound Power Level per unit, dBA
	AC-2-2	Daikin	FXMQ250PV1A	Workspace (north), workspace (east)	1	86
	AC-2-3	Daikin	FXMQ63PAVE	Meeting room 5, meeting room 6	1	80
	AC-2-4	Daikin	FXMQ140PAVE	Workspace (west)	1	85
	AC-3-1	Daikin	FXMQ25PAVE	First aid room	1	71
	AC-3-2	Daikin	FXMQ180PV1A	Workspace (west)	1	86
	AC-3-3	Daikin	FXMQ125PAVE	Training kitchen	1	82
	AC-3-4	Daikin	FXMQ125PAVE	Training room	1	82
	AC-3-5	Daikin	FXMQ125PAVE	South circulation	1	82
	AC-3-6	Daikin	FXMQ250PV1A	Dining	1	86
	AC-3-7	Daikin	FXMQ250PV1A	Dining	1	86
	AC-3-8	Daikin	FXMQ125PAVE	Board room	1	82
Exhaust Fan	EF-1	-	-	Amenities (Male, Female, PWD, Shower)	1	64

The mechanical plant layout has been based on the mechanical drawings by JHA Engineers.

For the purposes of this assessment, it has been assumed that the mechanical plant units (excluding ductwork) are enclosed within a ceiling plant area and installed above 16mm compressed fibre cement.

3.7.1. MECHANICAL EQUIPMENT NOISE RECOMMENDATIONS

The following recommendations are provided for the mechanical sub-contractor to ensure that noise from mechanical equipment does not exceed levels shown in Table 1.

3.7.1.1. Fan Coil Units

The following units in Table 13 have been identified to exceed the cumulative recommended internal design sound levels based on the current design and equipment selections. Recommended mitigation measures have been provided to reduce noise impacts.

Table 13: Recommended FCU mitigation measures

FCU	AREA SERVING	RECOMMENDATIONS
AC-1-1	Workspace (north)	• Supply – Install minimum 50mm thick, 9m long internal duct lining – Install minimum two (2) internally lined radiused duct bends and one (1) mitred duct bend – Supply spigot takeoff to be installed following 9m internally lined ductwork and three (3) internally lined duct bends (2 radiused bends and 1 mitred duct bend) – Install minimum 1m long flexible ductwork

FCU	AREA SERVING	RECOMMENDATIONS
		• Return – Install minimum 50mm thick internal lining within return plenum – Install minimum 50mm thick internal lining within the entire length of the return ductwork
AC-1-2	Meeting room 2, meeting room 4, private office	• Supply towards meeting room 2 – Install minimum 50mm thick, 2.5m long internal duct lining – Install minimum 50mm thick internal lining within supply plenum – Supply spigot takeoff to be installed following 2.5m internally lined ductwork – Minimum 1.5m of the supply duct length to be outside of the enclosed ceiling plant platform area – Supply grille to be connected by rigid duct. Avoid installing flexible ducts towards the supply – Wrap entire length of supply duct within the enclosed plant room with minimum 8kg/m^2 acoustic pipe lag (e.g., Pyrotek Soundlag 8025C) • Supply towards meeting room 4 – Install minimum 25mm thick, 0.5m long internal duct lining – Install minimum 50mm thick internal lining within supply plenum – Supply spigot takeoff to be installed following 0.5m internally lined ductwork – Supply grille to be connected by rigid duct. Avoid installing flexible ducts towards the supply – Wrap entire length of supply duct within the enclosed plant room with minimum 8kg/m^2 acoustic pipe lag (e.g., Pyrotek Soundlag 8025C) • Return from meeting rooms 2 and 4 – Install minimum 50mm thick internal lining within return plenum – Install minimum 50mm thick internal lining within the entire length of the duct run
AC-1-3	Workspace (north), workspace (east)	• Supply towards private office room – Install minimum 50mm thick, 4m long internal duct lining – Install minimum 50mm thick internal lining within supply plenum – Supply spigot takeoff to be installed following 2m internally lined ductwork – Minimum 4m of the supply duct length to be outside of the enclosed ceiling plant platform area – Supply grille to be connected by rigid duct. Avoid installing flexible ducts towards the supply • Supply towards workspace (north) – Install minimum 50mm thick, 7.5m long internal duct lining – Supply spigot takeoff to be installed following 7.5m internally lined ductwork – Install minimum 1.5m long flexible ductwork – At minimum, wrap the first 3m of the supply ductwork with minimum 4.5kg/m^2 acoustic pipe lag (e.g., Pyrotek Soundlag 4512) • Return from workspace (east) – Install minimum 50mm thick internal lining within return plenum – Install minimum 50mm thick internal lining within the entire length of the duct run
AC-1-4	Workspace (east)	• Supply towards workspace (east) – Install minimum 50mm thick, 11m long internal duct lining – Install minimum two (2) internally lined radiused duct bends – Supply spigot takeoff to be installed following 11m internally lined ductwork and two (2) internally lined duct bends – Install minimum 0.5m long flexible ductwork • Return from workspace (east) – Install minimum 50mm thick internal lining within return plenum

FCU	AREA SERVING	RECOMMENDATIONS
		– Install minimum 50mm thick internal lining within the entire length of the duct run
AC-1-5	Meeting room 3	• Supply – Install minimum 50mm thick, 5m long internal duct lining – Install minimum one (1) internally lined radiused duct bend – Supply spigot takeoff to be installed following 5m internally lined ductwork and one (1) internally lined duct bend – Supply grille to be connected by rigid duct. Avoid installing flexible ducts towards the supply – Wrap entire length of supply duct within the enclosed plant room with minimum 8kg/m^2 acoustic pipe lag (e.g., Pyrotek Soundlag 8025C) as illustrated in Figure 4 • Return – Install minimum 50mm thick internal lining within return plenum – Install minimum 25mm thick internal lining within the entire length of the duct run
AC-1-6	Meeting room 1	• Supply – Install minimum 50mm thick, 3m long internal duct lining – Install minimum one (1) internally lined radiused duct bends – Supply spigot takeoff to be installed following 3m internally lined ductwork and one (1) internally lined duct bends – Supply grille to be connected by rigid duct. Avoid installing flexible ducts towards the supply • Return – Install minimum 50mm thick internal lining within return plenum – Install minimum 25mm thick internal lining within the entire length of the duct run
AC-2-1	Workspace (north), workspace (west)	• Supply towards workspace (north) – Install minimum 50mm thick, 4.5m long internal duct lining – Install minimum one (1) internally lined radiused duct bend – Supply spigot takeoff to be installed following 4.5m internally lined ductwork and one (1) internally lined duct bends – Install minimum 2m long flexible ductwork – At minimum, wrap the first 2m of the supply ductwork with minimum 4.5kg/m^2 acoustic pipe lag (e.g., Pyrotek Soundlag 4512) as illustrated in Figure 4 • Supply towards workspace (west) – Install minimum 50mm thick, 11m long internal duct lining – Install minimum one (1) internally lined radiused duct bend – Supply spigot takeoff to be installed following 11m internally lined ductwork and one (1) internally lined duct bend – Install minimum 0.5m long flexible ductwork • Return from workspace (west) – Install minimum 50mm thick internal lining within return plenum – Install minimum 50mm thick internal lining within the entire length of the straight duct
AC-2-2	Workspace (north), workspace (east)	• Supply towards workspace (north) – Install minimum 50mm thick, 9.5m long internal duct lining – Install minimum two (2) internally lined radiused duct bend – Supply spigot takeoff to be installed following 9.5m internally lined ductwork and two (2) internally lined duct bends

FCU	AREA SERVING	RECOMMENDATIONS
		– Install minimum 2m long flexible ductwork • Return from workspace (west) – Install minimum 50mm thick internal lining within return plenum – Install minimum 3.5m long straight duct with 50mm internal duct lining – Install minimum two (2) internally lined radiused duct bends – Return inlet to be installed following 3.5m internally lined ductwork and two (2) internally lined duct bends
AC-2-3	Meeting room 5, meeting room 6	• Supply towards meeting room 5 – Install minimum 25mm thick, 2m long internal duct lining – Install minimum two (2) internally lined radiused duct bends – Supply spigot takeoff to be installed following 2m internally lined ductwork and two (2) internally lined duct bends – Supply grille to be connected by rigid duct. Avoid installing flexible ducts towards the supply – Wrap entire length of supply duct within the enclosed plant room with minimum 4.5kg/m² acoustic pipe lag (e.g., Pyrotek Soundlag 4512) • Supply towards meeting room 6 – Install minimum 25mm thick, 4m long internal duct lining – Install minimum one (1) internally lined radiused duct bend – Supply spigot takeoff to be installed following 4m internally lined ductwork and one (1) internally lined duct bends – Supply grille to be connected by rigid duct. Avoid installing flexible ducts towards the supply – Wrap entire length of supply duct within the enclosed plant room with minimum 4.5kg/m² acoustic pipe lag (e.g., Pyrotek Soundlag 4512) • Return – Install minimum 50mm thick internal lining within return plenum – Install minimum 25mm thick internal lining within the entire length of the duct run
AC-2-4	Workspace (west)	• Supply – Install minimum 50mm thick, 5m long internal duct lining – Supply spigot takeoff to be installed following 5m internally lined ductwork – Install minimum 2m long flexible ductwork • Return – Install minimum 50mm thick internal lining within return plenum – Install minimum 50mm thick internal lining within the entire length of the straight duct
AC-3-1	First aid room	• Supply – Install minimum 50mm thick, 1m long internal duct lining – Supply spigot takeoff to be installed following 1m internally lined ductwork – Install minimum 1.5m long flexible ductwork • Return – Install minimum 50mm thick internal lining within return plenum – Install minimum 2m long flexible ductwork
AC-3-2	Workspace (west)	• Supply – Install minimum 50mm thick, 7m long internal duct lining – Install minimum one (1) internally lined radiused duct bends

FCU	AREA SERVING	RECOMMENDATIONS
		– Supply spigot takeoff to be installed following a minimum of 7m internally lined ductwork and one (1) internally lined duct bend – Install minimum 0.5m long flexible ductwork – Wrap entire length of supply duct above the training kitchen with minimum 4.5kg/m^2 acoustic pipe lag (e.g., Pyrotek Soundlag 4512) • Return – Install minimum 50mm thick internal lining within return plenum – Install minimum 50mm thick internal lining within the entire length of the duct run
AC-3-3	Training kitchen	• Supply – Install minimum 50mm thick, 7.5m long internal duct lining – Install minimum one (1) internally lined radiused duct bend – Supply spigot takeoff to be installed following 7.5m internally lined ductwork and one (1) internally lined duct bend – Install minimum 1m long flexible ductwork – Wrap entire length of supply duct above the training kitchen with minimum 4.5kg/m^2 acoustic pipe lag (e.g., Pyrotek Soundlag 4512) • Return – Install minimum 50mm thick internal lining within return plenum – Install minimum 0.5m long flexible ductwork – Install minimum 50mm thick internal lining within the entire length of the duct run
AC-3-4	Training room	• Supply – Install minimum 50mm thick, 3m long internal duct lining – Install one (1) internally lined radiused duct bend and one (1) internally lined mitred duct bend – Supply spigot takeoff to be installed following 3m internally lined ductwork and two (2) internally lined duct bends – Supply grille to be connected by rigid duct. Avoid installing flexible ducts towards supply – Wrap entire length of supply duct within the enclosed plant room with minimum 8kg/m^2 acoustic pipe lag (e.g., Pyrotek Soundlag 8025C) • Return – Install minimum 50mm thick, 1.5m long internal duct lining – Install minimum 50mm thick internal lining within return plenum – Install one (1) internally lined radiused duct bend – Return inlet to be installed following 1m internally lined ductwork and one (1) internally lined duct bends – Install minimum 50mm thick internal lining within the entire length of the duct run
AC-3-5	South Circulation	• Supply – Install minimum 50mm thick, 4m long internal duct lining – Install one (1)internally lined radiused duct bend – Supply spigot takeoff to be installed following 4m internally lined ductwork and one (1) internally lined duct bend – Install minimum 0.5m long flexible ductwork • Return – Install minimum 50mm thick internal lining within the entire length of the duct run
AC-3-6	Dining	• Supply

<table>
<tr><th>FCU</th><th>AREA SERVING</th><th>RECOMMENDATIONS</th></tr>
<tr><td></td><td></td><td>– Install minimum 50mm thick, 13m long internal duct lining
– Supply spigot takeoff to be installed following 13m internally lined ductwork
– Install minimum 1m long flexible ductwork
• Return
– Install minimum 50mm thick internal lining within return plenum
– Install minimum 50mm thick internal lining within the entire length of the duct run
–</td></tr>
<tr><td>AC-3-7</td><td>Dining</td><td>• Supply
– Install minimum 50mm thick, 13m long internal duct lining
– Supply spigot takeoff to be installed following 13m internally lined ductwork
– Install minimum 1m long flexible ductwork
• Return
– Install minimum 50mm thick internal lining within return plenum
– Install minimum 50mm thick internal lining within the entire length of the duct run
–</td></tr>
<tr><td>AC-3-8</td><td>Board room</td><td>• Supply
– Install minimum 50mm thick, 6m long internal duct lining
– Install one (1) internally lined radiused duct bend
– Supply spigot takeoff to be installed following 6m internally lined ductwork and one (1) internally lined duct bend
– Install minimum 1m long flexible ductwork
• Return
– Install minimum 50mm thick internal lining within return plenum
– Install minimum 50mm thick internal lining within the entire length of the duct run</td></tr>
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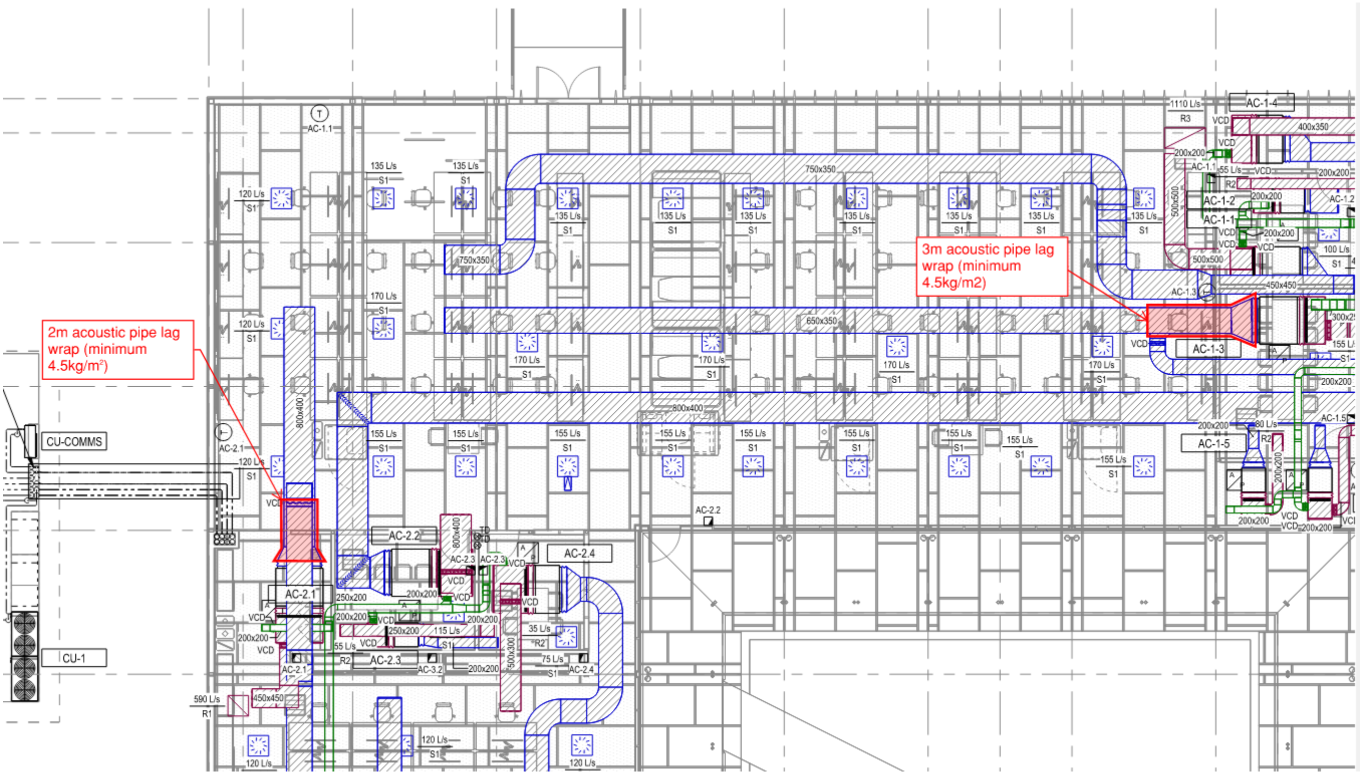


Figure 4: Open office workspace supply duct acoustic pipe lag markup

3.7.1.2. Fans

In general, allowance should be made for the following:

- In-ceiling fans shall be wrapped in 10kg/m^2 mass loaded vinyl

The following units in Table 14 have been identified to exceed the cumulative recommended internal design sound levels based on the current design and equipment selections. Recommended mitigation measures have been provided to reduce noise impacts.

Table 14: Recommended fan mitigation measures

FCU	AREA SERVING	RECOMMENDATIONS
EF-1	Amenities (Male, Female, PWD, Shower)	• Install minimum 1.5m long flexible ductwork

3.7.2. VIBRATION ISOLATION

All mechanical plant equipment shall be fitted with vibration isolation in accordance with manufacturers specifications.

3.7.3. SYSTEM GENERATED NOISE

The following recommendations are provided for the mechanical sub-contractor to ensure that generated noise within the air distribution system does not exceed levels shown in Table 1. The sub-contractor shall ensure that a minimum of air borne noise generated by air turbulence, air leakage, air moving equipment or by any other mechanical services equipment, is transmitted through air ducts, clearance holes etc.

3.7.3.1. Penetrations

The sub-contractor shall seal around pipes, ducts, cables and any other services, which they have provided, where these pass through walls, ceilings and floors. All sealing details utilised shall be designed to maintain the acoustic rating of all walls and ceilings, which they penetrate.

3.7.3.2. Terminal Units

Constant volume and variable volume terminal boxes, induction units etc. shall be selected to ensure that their self-generated down duct and radiated noise, together from noise from the main system which is transmitted through the duct work, does not cause required space noise levels to be exceeded under normal condition of operation. The down duct and radiated sound power levels for terminal units at the operating conditions shall be submitted with the Tender.

3.7.3.3. Dampers

Branch dampers shall be used for balancing with only minor adjustments made to diffuser faces. Where damping of diffusers indicates an oversupply of air, the fan speed shall be reduced, or the terminal box, (e.g. mixing box) shall be re-adjusted. Air distribution systems, which are noisy due to excessive damping, shall be re-balanced.

3.7.3.4. Duct Air Velocities

Duct air velocities with in ducts and risers shall not exceed those listed in Table 15 below.

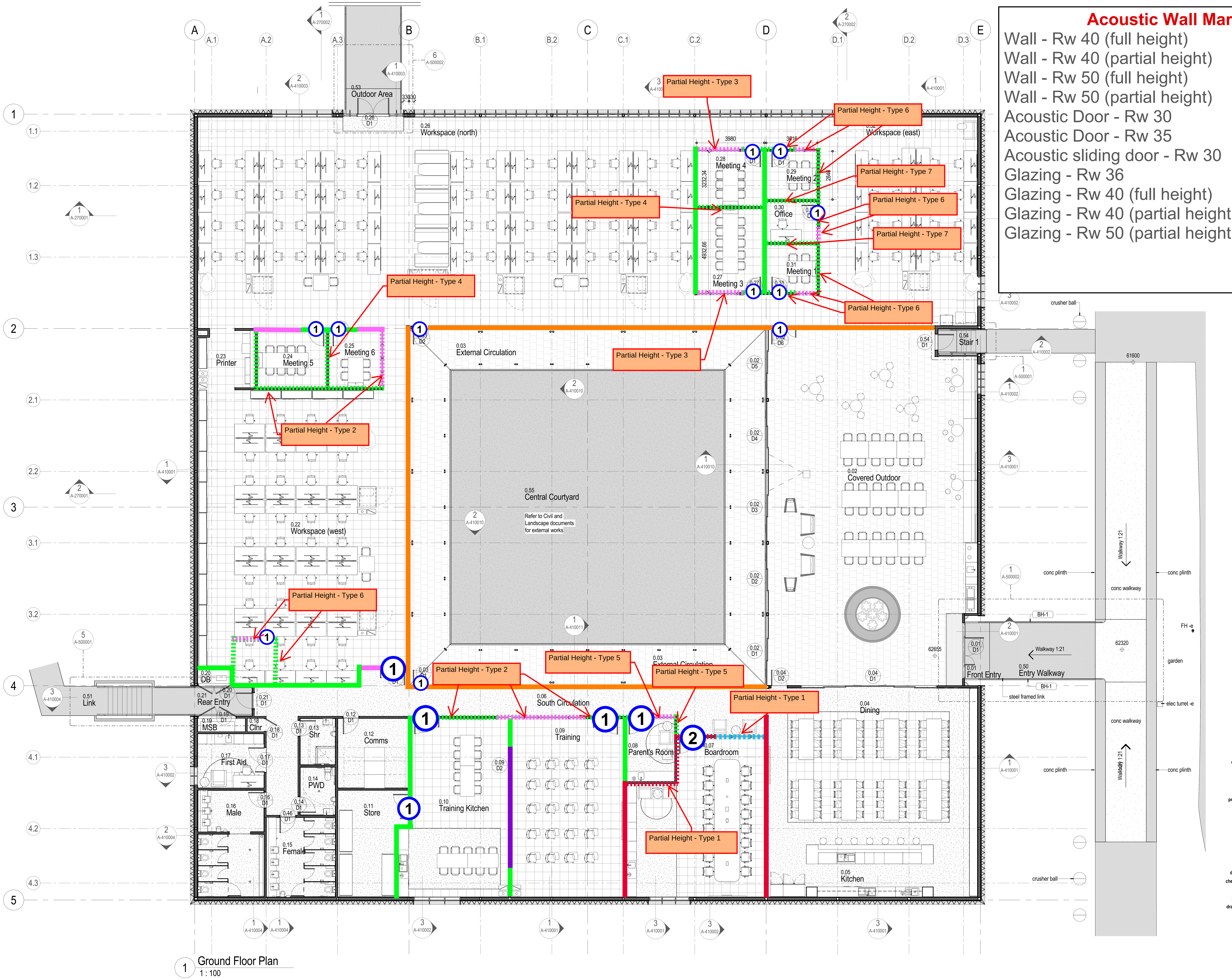
Table 15: Maximum duct velocities

Noise criterion, dBA	Maximum Duct Velocities, m/s				
	Main riser	Primary branch	Secondary riser	Run outs	Flexible
30	8 – 10	5.8	4.5	2.9	1.9
35	10 – 12	6.9	5.5	3.6	2.5
40	12 – 15	8.4	6.8	4.3	2.9
45	15 – 18	10.2	8.2	5.2	3.6
50	18 – 23	12.5	10.0	6.5	4.3
55	23 – 29	15	12.0	8.0	5.2

Appendix A: Architectural Drawings

DRAWING NO.	REVISION	TITLE	DATE ISSUED
A-200001[10]	11	General Arrangement Plan	28/11/2025
A-210001[9]	9	Reflected Ceiling Plan	28/11/2025
A-220001[5]	5	Wall Types	28/11/2025
A-220002[5]	5	Wall Types Roof Cavity	28/11/2025
A-230001[5]	5	Floor Finishes Plan	28/11/2025
A-260001[9]	9	General Elevations	28/11/2025
A-260002[9]	9	General Elevations	28/11/2025
A-270001[9]	9	General Sections	28/11/2025
A-270002[7]	7	General Sections	28/11/2025
A-600001 to A-600017	-	Interior Plan and Elevations	28/11/2025

Appendix B: Internal Wall Mark-up



Ground Floor Plan
1 : 100

Acoustic Wall Mark-up

- Wall - Rw 40 (full height)
- Wall - Rw 40 (partial height)
- Wall - Rw 50 (full height)
- Wall - Rw 50 (partial height)
- Acoustic Door - Rw 30 (1)
- Acoustic Door - Rw 35 (2)
- Acoustic sliding door - Rw 30
- Glazing - Rw 36
- Glazing - Rw 40 (full height)
- Glazing - Rw 40 (partial height)
- Glazing - Rw 50 (partial height)

Rev 4 - 25.11.2025

Drawings include colour information. A black & white copy may not have all the information. This background is coloured blue.

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project Stanwell New Projects Office
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title General Arrangement Plan

drawn bg
checked jb
date Sept 2025
scale 1 : 100 at A1
drawing 25010 / dd A-200001 / 6

1:100 at A1
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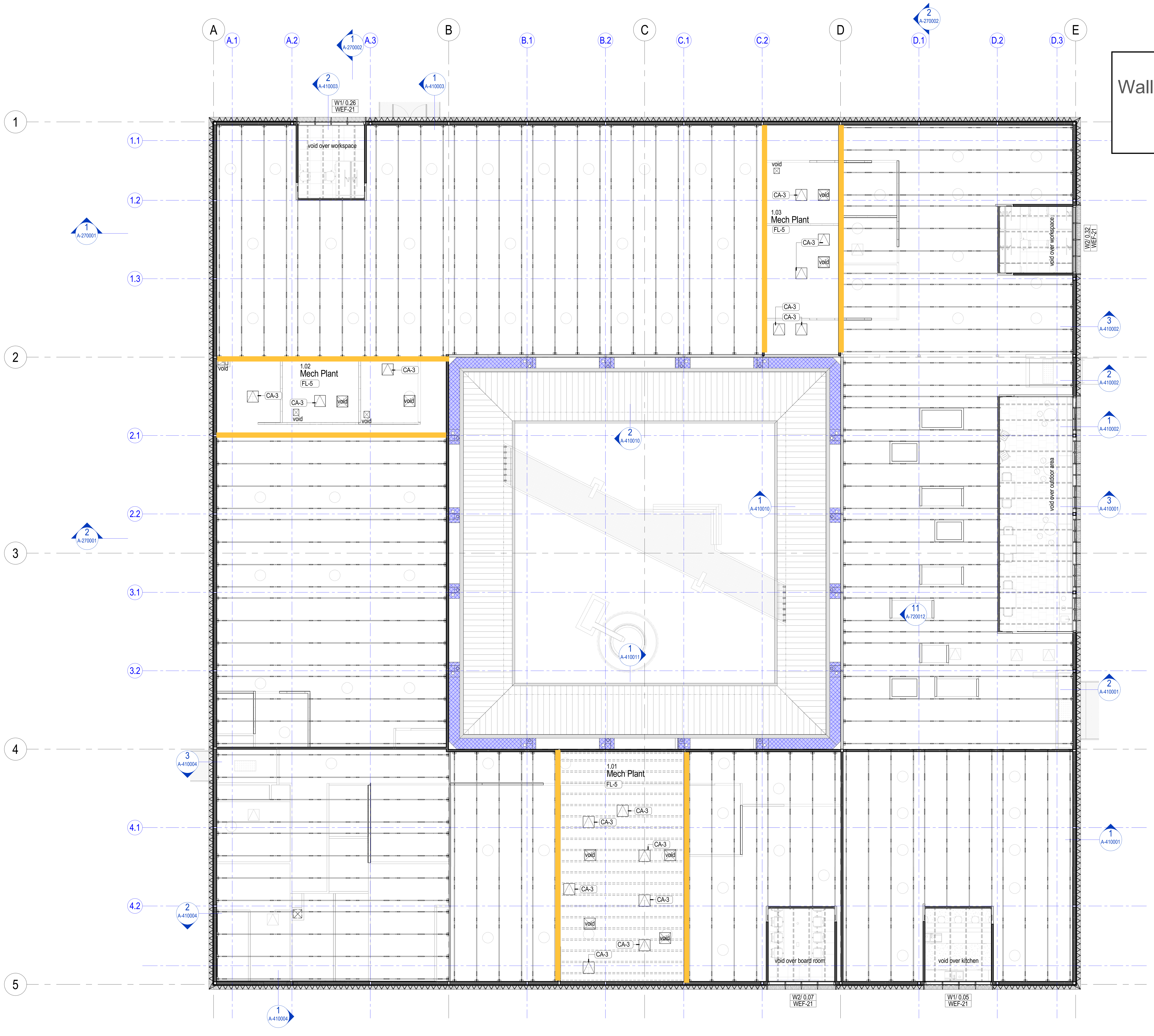

1 Roof Space Plan
1 : 100

Acoustic Wall Mark-up

Wall - Rw 35 (full height)

Rev 4 - 25.11.2025

ROOF NOTES

RN.01 UNO all roof flashings to match roof sheeting.

RN.02 Provide fall arrest system to relevant australian standards. Provide a

RN.05 Refer also to Mechanical and Hydraulic Services documents for roof penetrations.

rev	date	description	initial
1	17/06/25	QS Issue #1	
2	08/07/2025	Background Issue	
3	11/08/2025	Coordination Issue	JB
4	29/08/2025	Coordination Issue	JB
5	08/09/2025	QS Issue	JB
		D Issue	JB
		D Issue	JB
		ordination Issue	JB

ROOF LEGEND

box gutter sump - refer to hydraulic docs for performance based brief

PRELIMINARY
NOT FOR CONSTRUCTION

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title Roof Cavity Plan

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checked jb
date Nov 2025
scale 1 : 100 at A1
drawing 25010 / cd A-200003 / 8

1:100 at A1
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Appendix C: Glossary of Acoustic Terminology

Decibel – dB – relative unit of measurement for acoustic power, pressure and intensity defined by the ratio of square of the sound pressure, power or intensity to a reference sound pressure, power, or intensity value (usually the threshold of human hearing at 1 kHz). Any value expressed as "level" will use decibels as units. Humans have a large sound-sensitivity range, so values are expressed in decibels for a more practical range. Values expressed in decibels such as sound pressure level and sound power level cannot be added arithmetically, as their pressure or power values are expressed as a logarithmic ratio. Two equal sound levels combined will result in a sound pressure level of 3dB higher than the sound level of one source (e.g., 60 dB + 60 dB = 63 dB). Levels with 10 or more dB difference will not be added (e.g., 50 dB + 60 dB = 60 dB). All values in this report expressed in decibels assume reference pressure of 20 µPa.

A-weighted decibel – dB(A), dBA – frequency weighted sound levels in decibels correlated with perceived human hearing at low and medium levels. dB(A) and dBA are used to express the units; A used as a subscript e.g., L_{Aeq} or L_{A90} denotes an A-weighting applied to that value.

Sound Pressure Level – SPL, L – sound pressure measured in decibels. Logarithmic values relative to a reference value are used to convert the large range of sound pressure (in Pascals) audible to humans to a more practical range. Sound pressure level is a measured value and is dependent on distance from the sound source(s) and acoustic environment.

Sound Power Level – SWL – sound power in decibels. Logarithmic values relative to a reference value are used to convert the large range of sound power (in Watts) audible to humans to a more practical range. Sound power level is a calculated value that is inherent to a sound source and is independent of distance and acoustic environment.

Reverberation – the accumulation of sound energy caused by the reflection of sound on the surfaces in a room.

Reverberation time – RT, RT60, T60 – the time it takes sound in a room to decay 60dB after the sound source has stopped. T60 is often extrapolated from T30 or T20 where the background noise does not allow 60dB of decay to be measured. RT is dependent on room volume and amount of acoustically absorptive surfaces. RT is measured in seconds.

Octave band (and centre frequency) – octave bands divide the spectrum of audible sound into equal parts. An octave band is denoted by its "centre frequency," in Hertz, Hz. Each octave or octave band includes a range of frequencies whose upper frequency limit is twice that of its lower frequency limit. For example, the 1000 Hz octave band contains sound energy at all frequencies from 707 Hz to 1414 Hz, rounded to 710 Hz and 1410 Hz for practical reasons. One-third octave bands span one-third of an octave and are often used for more precise applications.

L_{eq} or $L_{eq,T}$ – The equivalent continuous sound level is the energy average of the varying noise over the sample period (often specified in the subscript) and is equivalent to the level of a constant noise which contains the same energy as the varying noise environment. This measure is also a common measure of environmental noise and road traffic noise. L_{eq} is measured in dB.

L_{Aeq} or $L_{Aeq,T}$ – A-weighted L_{eq} measured in dB(A).

L_{90} or $L_{90,T}$ – The noise level which is exceeded for 90% of the sample period. During the sample period, the noise level is below the L_{90} level for 10% of the time. This measure is commonly referred to as the background noise level or RBL. L_{90} is measured in dB.

L_{A90} or $L_{A90,T}$ – A-weighted L_{90} measured in dB(A).

NRC – a single number index from 0 to 1 representing the proportion of sound energy reflected by a surface where 1 equals total absorption and 0 equals zero absorption (total reflection), given by the average value of the absorption coefficients in the 250 Hz, 500 Hz, 1 kHz, and 2 kHz octave bands. NRC is used to compare sound absorption performance in the midrange of frequencies for building materials.

Acoustic insulation – a general term to describe the ability or effectiveness of a building element such as a wall, window, door, or floor to reduce sound transmission depending on its composition and construction. Insulation materials such as fiberglass and polyester – often referred to as "insulation" – can be used in walls, floors, ceilings etc. to reduce interstitial reflections in the cavity which may increase the acoustic insulation performance.

Transmission loss – the reduction in sound pressure level between two designated locations in a sound transmission system.

Sound insulation transmission loss – the difference in decibels between the average sound pressure levels across a partition separating source and receiver rooms.

R_w – Weighted Sound Reduction Index – the design value representing the effective sound reduction of a building element. Each increasing increment in R_w is equivalent to 1 dB of noise reduction. R_w is based on laboratory measurement, where negligible flanking is present. Spectrum adaptation terms C and C_{tr} are often added to the measured R_w result to account for low frequency noise. R_w is measured in (linear) dB.

$L_{n,w}$ – Weighted Normalised Impact Sound Pressure Level – the design value of the achievable impact noise attenuation of a building element. $L_{n,w}$ quantifies the perceived impact noise in the receiver room, with lower values corresponding to lower levels of theoretical perceived impact noise. Each increasing increment in $L_{n,w}$ is equivalent to 1 dB of impact noise increase. $L_{n,w}$ is based on laboratory measurement, where negligible flanking is present. Spectrum adaptation term C_i is often added to the $L_{n,w}$ result to account for low frequency noise. $L_{n,w}$ is measured in (linear) dB.

Speech privacy – how intelligible an overheard conversation is to a subject given by difference level of intruding sound. This can be in terms of conversations in the same space, or a conversation happening on the opposite side of a partition. Speech privacy depends upon background noise level, the reverberation time of the space and the airborne sound insulation of a partition. Note that even with high quality acoustic privacy and low levels of intelligibility, keen eavesdroppers may still be able to understand speech from adjoining rooms, if they listen from a position close to the adjoining partition (mostly due to the English language containing a significant number of redundant words).

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